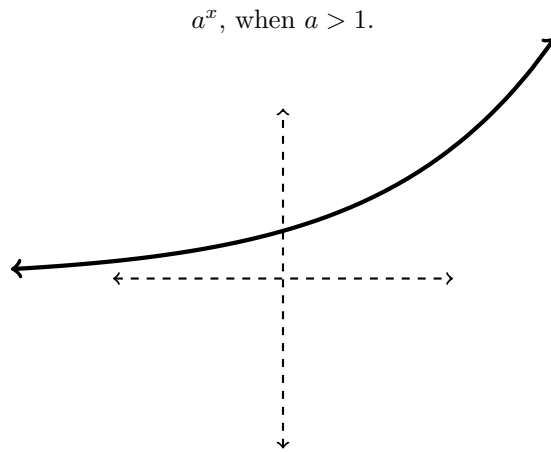


MATH 195: 9/29
EXPONENTIAL FUNCTIONS

An *exponential function* is one you can write in the form

$$f(x) = a^x,$$

where $a \neq 1$ is a positive number.



Horizontal asymptote at $y = 0$. Increasing everywhere and concave up everywhere.

Like with other functions, it is useful to understand geometric transformations of exponential functions. When graphing one, the two things you need to determine are (1) its orientation and (2) where the horizontal asymptote is.

Some of these exercises ask you to use a graphing calculator, such as the one found at [desmos.com/calculator](https://www.desmos.com/calculator).

- (1) Sketch a graph of and explain why $a(x) = 2 - 2^x$ has its horizontal asymptote at $y = 2$ and has range $(-\infty, 2)$. Check your work using a graphing calculator.
- (2) Sketch graphs of all four orientations for an exponential function with base, say, 3. Write formulas for each, and check your work by graphing those functions with a graphing calculator.
- (3) Your friend claims that it is impossible for a geometric transformation of an exponential function to be both increasing and concave down. Prove them wrong. Which combinations of increasing versus decreasing and concave up versus concave down are possible?
- (4) Use a graphing calculator to graph both $b(x) = \left(\frac{1}{2}\right)^x$ and $c(x) = 2^{-x}$. Explain what you see. Generalize your observation to explain how graphs of exponential functions a^x with $0 < a < 1$ must look.
- (5) Use a graphing calculator to graph both $u(x) = 4 \cdot 2^x$ and $v(x) = 2^{x+2}$. Explain what you see. Generalize your observation to explain the relationship between vertical scalings and horizontal translations of exponential functions.
- (6) Use a graphing calculator to graph both $q(x) = 3^{2x}$ and $r(x) = 9^x$. Explain what you see. Generalize your observation to explain what a horizontal scaling of an exponential function does.
- (7) Using a graphing calculator to help you, write down a formula for an exponential function f satisfying all of the following: (1) it is decreasing everywhere, (2) the range is $(-4, \infty)$, and (3) $f(0) = 0$.
- (8) Given any possible range $(-\infty, a)$ or (a, ∞) for an exponential function and any value b in that range, how do you find an exponential function g with that range with $g(0) = b$? Can you come up with a general formula based on a and b ?
- (9) Use a graphing calculator to compare power functions and exponential functions to see which one grows faster as $x \rightarrow \infty$. Can you observe a general pattern? State a conjecture about how to tell whether x^n or a^x grows faster.
- (10) Use a graphing calculator to compare reciprocals of power functions and exponential functions to see which one goes to 0 faster as $x \rightarrow -\infty$. Can you observe a general pattern? State a conjecture about how to tell whether x^{-n} or a^x goes to 0 faster.